



Containerized bioinformatics applications and pipelines on Open OnDemand

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alphafold

alphafold

- AlphaFold is an AI system developed by DeepMind that has transformed the field of structural biology. It accelerates research in drug discovery, biotechnology, and fundamental biology.
- The 2024 Nobel Prize in Chemistry was awarded to John Jumper, Demis Hassabis, and David Baker for developing AlphaFold, an AI tool that predicts protein structures.



Challenges of Managing AlphaFold for System Administrators

full_db_20230311.ff (for alphafold v2.3)

Example contents of full_db_20231031.ff for monomer:

```
--db_preset=full_dbs
--bfd_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/bfd/bfd_metaclust_clu_co
--data_dir=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/
--uniref90_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/uniref90/uniref90.f
--mgnify_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/mgnify/mgy_clusters_2
--uniref30_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/uniref30/UniRef30_2
--pdb70_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb70/pdb70
--template_mmcif_dir=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb_mmcif/mmcif_files
--obsolete_pdbs_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb_mmcif/obsolete.dat
--hhblits_binary_path=/usr/bin/hhblits
--hhsearch_binary_path=/usr/bin/hhsearch
--jackhmmer_binary_path=/usr/bin/jackhmmer
--kalign_binary_path=/usr/bin/kalign
```

Example contents of full_db_20231031.ff for multimer:

```
--db_preset=full_dbs
--bfd_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/bfd/bfd_metaclust_clu_co
--data_dir=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/
--uniref90_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/uniref90/uniref90.f
--mgnify_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/mgnify/mgy_clusters_2
--uniref30_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/uniref30/UniRef30_2
--pdb_seqres_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb_seqres/pdb_se
--uniprot_database_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/uniprot/uniprot.fast
--template_mmcif_dir=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb_mmcif/mmcif_files
--obsolete_pdbs_path=/cluster/tufts/biocontainers/datasets/alphafold/db_20231031/pdb_mmcif/obsolete.dat
--hhblits_binary_path=/usr/bin/hhblits
--hhsearch_binary_path=/usr/bin/hhsearch
--jackhmmer_binary_path=/usr/bin/jackhmmer
--kalign_binary_path=/usr/bin/kalign
```

Challenges of Managing AlphaFold for System Administrators

Example job using GPU

To run alphafold using GPU:

```
#!/bin/bash
#SBATCH -p PartitionName # gpu or preempt
#SBATCH -t 24:00:00
#SBATCH -N 1
#SBATCH -n 1
#SBATCH -c 10
#SBATCH --mem=64G
#SBATCH --gres=gpu:1
#SBATCH --job-name=alphafold
#SBATCH --mail-type=FAIL,BEGIN,END
#SBATCH --error=%x-%J-%u.err
#SBATCH --output=%x-%J-%u.out

module purge
module load alphafold/2.3.2

run_alphafold.sh --flagfile=full_db_20231031.ff \
  --fasta_paths=sample.fasta --max_template_date=2022-02-01 \
  --output_dir=af2_full_out --model_preset=monomer \
  --use_gpu_relax=True
```

Example job using CPU

To run alphafold using CPU:

```
#!/bin/bash
#SBATCH -p PartitionName # batch or your group's own partition
#SBATCH -t 24:00:00
#SBATCH -N 1
#SBATCH -n 1
#SBATCH -c 10
#SBATCH --mem=64G
#SBATCH --job-name=alphafold
#SBATCH --mail-type=FAIL,BEGIN,END
#SBATCH --error=%x-%J-%u.err
#SBATCH --output=%x-%J-%u.out

module purge
module load alphafold/2.3.2

run_alphafold.sh --flagfile=full_db_20231031.ff \
  --fasta_paths=sample.fasta --max_template_date=2022-02-01 \
  --output_dir=af2_full_out --model_preset=monomer \
  --use_gpu_relax=False
```

AlphaFold2 on OOD

Interactive Apps
AI
 ComfyUI
 Ollama + Jupyter
 Ollama + Open WebUI
 Stable Diffusion
 text-generation-webui
Admin tools
 Research Technology Guides
Apps
 FastQC
 Fiji
Bioinformatics
 AlphaFold
 AlphaFold3
Classroom
 Jupyter for classroom
Servers
 Jupyter
 Matlab server
 RStudio Server
 VSCode Server

nf-core pipelines
atacseq
 atacseq 2.1.2
chipseq
 chipseq 2.1.0
genomics
 pangenome 1.1.2

AlphaFold

This app will launch [AlphaFold](#) on [Tufts HPC](#).

Number of hours

Number of cores

Select the number of CPU cores for your job.

Amount of Memory (GB)

Enter the memory amount in GB (e.g., **12** for 12GB or **100** for 100GB).

Partition

Jobs submitted to the 'preempt' partition may be automatically killed to allow higher-priority jobs to run. Select a specific lab partition if you belong to that group.

Software Version

Database

Working directory

Select Path

Output directory Name

Where the results will be going to (relative to the working directory field above).
Example: alphafold.out

fasta_paths

AlphaFold3: More Powerful, Yet More Complex for Users

nature

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[nature](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 08 May 2024

Accurate structure prediction of biomolecular interactions with AlphaFold 3

[Josh Abramson](#), [Jonas Adler](#), [Jack Dunger](#), [Richard Evans](#), [Tim Green](#), [Alexander Pritzel](#), [Olaf Ronneberger](#), [Lindsay Willmore](#), [Andrew J. Ballard](#), [Joshua Bambrick](#), [Sebastian W. Bodenstein](#), [David A. Evans](#), [Chia-Chun Hung](#), [Michael O'Neill](#), [David Reiman](#), [Kathryn Tunyasuvunakool](#), [Zachary Wu](#), [Akvilė Žemgulytė](#), [Eirini Arvaniti](#), [Charles Beattie](#), [Ottavia Bertolli](#), [Alex Bridgland](#), [Alexey Cherepanov](#), [Miles Congreve](#), ... [John M. Jumper](#)  [+ Show authors](#)

[Nature](#) **630**, 493–500 (2024) | [Cite this article](#)

757k Accesses | **2007** Altmetric | [Metrics](#)

```
docker run -it \  
  --volume $HOME/af_input:/root/af_input \  
  --volume $HOME/af_output:/root/af_output \  
  --volume <MODEL_PARAMETERS_DIR>:/root/models \  
  --volume <DATABASES_DIR>:/root/public_databases \  
  --gpus all \  
  alphafold3 \  
  python run_alphafold.py \  
  --json_path=/root/af_input/fold_input.json \  
  --model_dir=/root/models \  
  --output_dir=/root/af_output
```

AlphaFold3 on OOD

Interactive Apps

AI

ComfyUI

Ollama + Jupyter

Ollama + Open WebUI

Stable Diffusion

text-generation-webui

Admin tools

Research Technology Guides

Apps

FastQC

Fiji

Bioinformatics

AlphaFold

AlphaFold3

Classroom

Jupyter for classroom

Servers

Jupyter

Matlab server

RStudio Server

VSCode Server

nf-core pipelines

atacseq

atacseq 2.1.2

chipseq

chipseq 2.1.0

genomics

pangenome 1.1.2

AlphaFold3

This app will launch AlphaFold3 developed by DeepMind.

Number of hours

2

Number of cores

8

Select the number of CPU cores for your job.

Amount of Memory (GB)

16000

Enter the memory amount in GB (e.g., 12 for 12GB or 100 for 100GB).

Partition

batch

Jobs submitted to the 'preempt' partition may be automatically killed to allow higher-priority jobs to run. Select a specific lab partition if you belong to that group.

Version

3.0.1

Working directory

/cluster/tufts/biocontainers/tests/alphafold3

Select your project directory; defaults to \$HOME

Select Path

Single input or multiple inputs

Single input

json_path

/cluster/tufts/biocontainers/tests/alphafold3/inputs/fold_input.json

Path to the input JSON file.

Select Path



Priority: P4 - Medium

Short description: [Tufts HPC] Installing R Packages

Description:

Hi there,

I am reaching out again on the topic of installing R packages because I'm struggling to install the package, "seqinr".

When I attempt to install, I get the following (long) errors:

```
> install.packages("seqinr")
```

```
Installing package into '/cluster/home/emosel01/R/x86_64-pc-linux-gnu-library/4.3'
```

```
(as 'lib' is unspecified)
```

```
Warning: dependency 'MASS' is not available
```

```
also installing the dependencies 'ade4', 'segmented'
```

Containerization: The Ultimate Solution for GLIBC Compatibility Issues

I have recently tried to install the R-package zellkonverter and have been running into the following error message:

```
* installing *source* package 'zellkonverter' ...
```

```
** using non-staged installation via StagedInstall field
```

```
Error in dyn.load(file, DLLpath = DLLpath, ...) :
```

```
unable to load shared object '/cluster/home/user01/R/x86_64-pc-linux-gnu-  
library/4.0/Matrix/libs/Matrix.so':
```

```
/lib64/libm.so.6: version `GLIBC_2.35' not found (required by  
/cluster/home/gperer01/R/x86_64-pc-linux-gnu-library/4.0/Matrix/libs/Matrix.so)
```

```
Calls: :: ... asNamespace -> loadNamespace -> library.dynam -> dyn.load
```

```
Execution halted
```

```
ERROR: configuration failed for package 'zellkonverter'
```



R-RNAseq

Customized R container for RNAseq analysis.

- ComplexHeatmap
- DESeq2
- DEXSeq
- edgeR
- ggrepel
- Limma
- pheatmap
- tidyverse



R-scRNAseq

Customized R container for scRNAseq analysis.

- CoGAPS
- DESeq2
- doSNOW
- DropletUtils
- edgeR
- miQC
- monocle
- monocle3
- Nebulosa
- scCATCH
- scDblFinder
- SCHNAPPs
- scMappR
- seurat
- seurat-wrappers
- SingleR
- SnapATAC
- SoupX
- velocity.R
- And more



tuftsttsrt/rserver

By [tuftsttsrt](#) · Updated 1 day ago

RStudio Server that has been deployed on Tufts HPC

[IMAGE](#)

DATA SCIENCE

★1 ↓ 94

Dockerfile



rocker/tidyverse

By [rocker](#) • Updated about 18 hours ago

Version-stable build of R, rstudio, and R packages

IMAGE

DATA SCIENCE

LANGUAGES & FRAMEWORKS

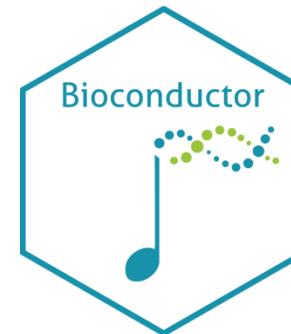
☆194 ⬇10M+

Enable
copilot

apt install
rpms

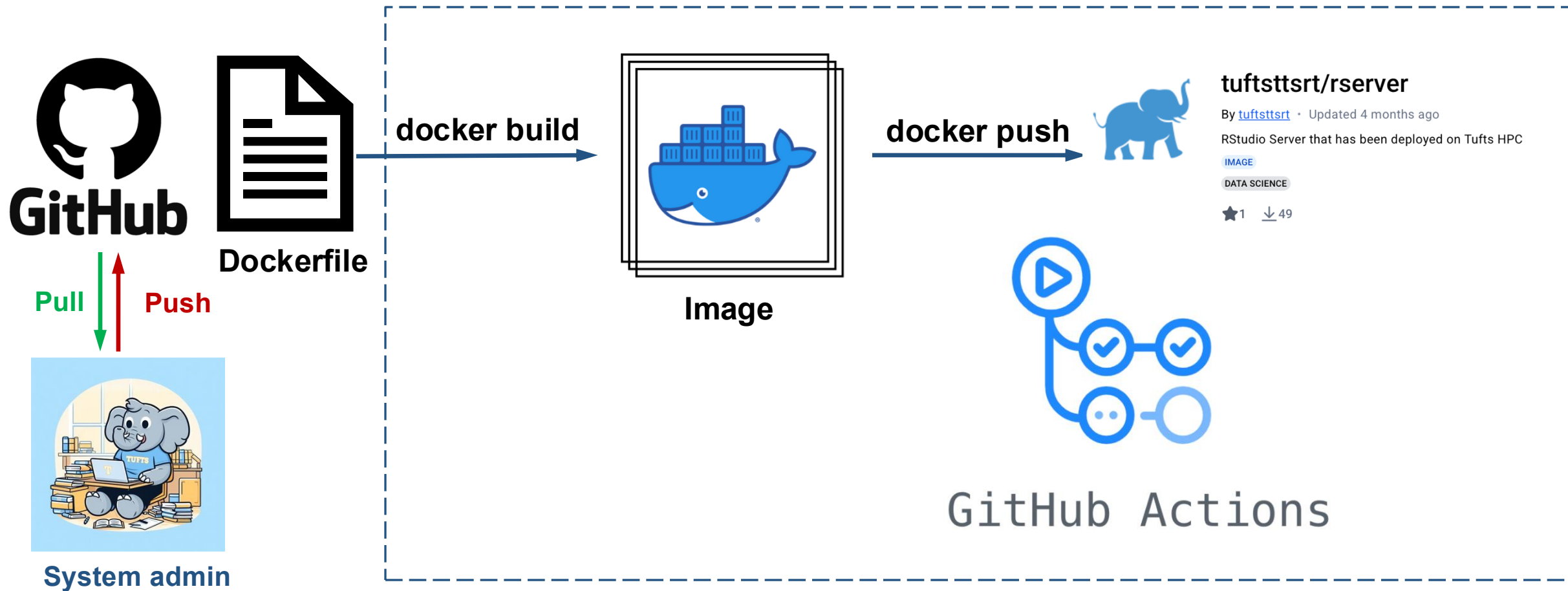
CMake,
FFTW &
dependencies


GitHub
packages



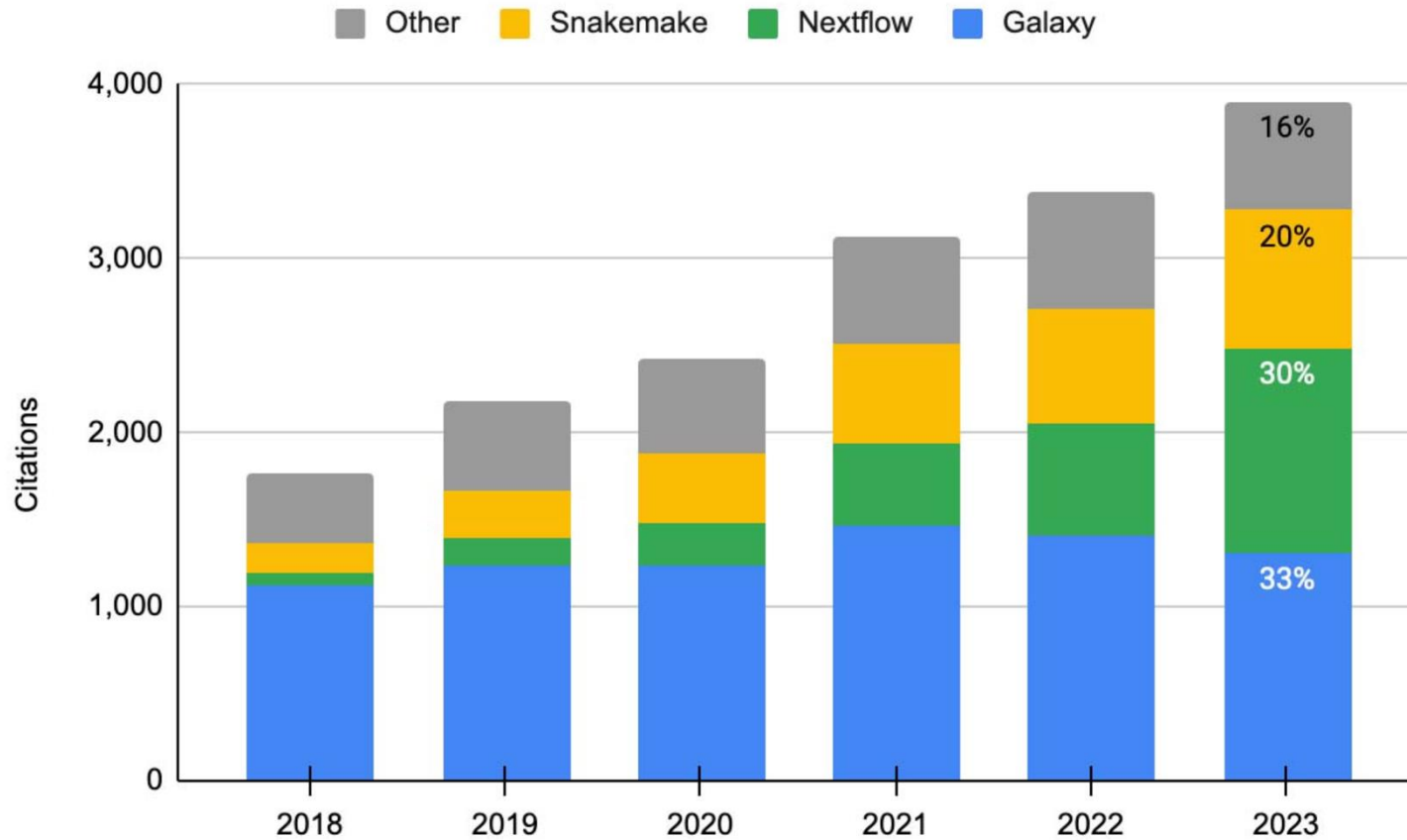
CRAN **R**
The Comprehensive R
Archive Network

Build & Manage container images using GitHub Actions & Docker Hub



Nextflow & nf-core

Nextflow Pipelines: The Future of Bioinformatics Workflows



Langer, Bjorn E., et al.
"Empowering bioinformatics communities with Nextflow and nf-core." *bioRxiv* (2024): 2024-05.

FIGURE 1: Google Scholar citation counts for bioinformatics workflow management systems. Sum of citations of the major publications of Galaxy, Nextflow, and Snakemake between 2018 and 2023 (Data in Supplementary Table 1).

nextflow pipeline

Write code
in any language



Orchestrate tasks with
dataflow programming



Define software
dependencies
via containers



Built-in version
control with Git



nextflow runtime

Task orchestration
and execution

Supported Platforms





A community effort to collect a curated set of analysis pipelines built using Nextflow.

Pipelines

Browse the 126 pipelines that are currently available as part of nf-core.

Released 78

Under development 36

Archived 12

Stars

Grid View

rnaseq ✓

☆ 988

RNA sequencing analysis pipeline using STAR, RSEM, HISAT2 or Salmon with gene/isoform counts and extensive quality control.

rna rna-seq

3.18.0 released 2 months ago

sarek ✓

☆ 424

Analysis pipeline to detect germline or somatic variants (pre-processing, variant calling and annotation) from WGS / targeted sequencing

annotation cancer gatk4 genomics germline
pre-processing somatic target-panels
variant-calling whole-exome-sequencing
whole-genome-sequencing

3.5.1 released 15 days ago

scrnaseq ✓

☆ 251

Single-cell RNA-Seq pipeline for barcode-based protocols such as 10x, DropSeq or SmartSeq, offering a variety of aligners and empty-droplet detection

10x-genomics 10xgenomics alevin bustools
cellranger kallisto rna-seq single-cell star-solo

3.0.0 released 3 months ago

mag ✓

☆ 227

Assembly and binning of metagenomes

annotation assembly binning long-read-sequencing
metagenomes metagenomics nanopore
nanopore-sequencing

3.3.1 released 20 days ago

chipseq ✓

☆ 206

ChIP-seq peak-calling, QC and differential analysis pipeline.

chip chip-seq chromatin-immunoprecipitation

atacseq ✓

☆ 197

ATAC-seq peak-calling and QC analysis pipeline

nanoseq ✓

☆ 195

Nanopore demultiplexing, QC and alignment pipeline

ampliseq ✓

☆ 193

Amplicon sequencing analysis workflow using DADA2 and QIIME2

16s 18s amplicon-sequencing edna illumina
iontorrent its metabarcoding metagenomics
microbiome pacbio qiime2 rrna

Institutional profiles

```
nextflow run nf-core/<pipeline> -profile mycluster
```

⇒ Specifies job submission

📦 Specify software packaging

Works for:

🏠 🧑‍🔬 🧪 For all pipelines

👤 👤 For all users on your system

🔄 Single point to update

tufts profile



SINGULARITYCE

```
params {  
    max_memory = 120.GB  
    max_cpus = 72  
    max_time = 168.h  
    igenomes_base = '/cluster/tufts/biocontainers/datasets/igenomes/'  
}
```

```
process {  
    executor = 'slurm'  
    clusterOptions = '-N 1 -n 1 -p batch'  
}
```

```
executor {  
    queueSize = 16  
    pollInterval = '1 min'  
    queueStatInterval = '5 min'  
    submitRateLimit = '10 sec'  
}
```

```
// Set $NXF_SINGULARITY_CACHEDIR in your ~/.bashrc  
// to stop downloading the same image for every run
```

```
singularity {  
    enabled = true  
    autoMounts = true  
}
```

<https://github.com/nf-core/configs/blob/master/conf/tufts.config>

```
#!/bin/bash
```

```
#SBATCH --time=00-48:00:00
```

```
#SBATCH -p batch
```

```
#SBATCH -N 1
```

```
#SBATCH -n 1
```

```
#SBATCH -c XX
```

```
#SBATCH --mem=XXG
```

```
#SBATCH --job-name nf-core
```

```
#SBATCH --output=%x-%J-%u.out
```

```
#SBATCH --error=%x-%J-%u.err
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-user=XXX@tufts.edu
```

```
module load nf-core/2.10
```

```
export NXF_SINGULARITY_CACHEDIR=/cluster/tufts/biocontainers/nf-core/singularity-images
```

```
nextflow run /cluster/tufts/biocontainers/nf-core/pipelines/nf-core-rnaseq/3.14.0/3_14_0/ \
```

```
    --input samplesheet.csv --outdir output \
```

```
    --fasta ref.fasta --gtf ref.gtf --aligner star_salmon \
```

```
    -profile singularity \
```

```
    --max_memory XXG --max_cpus XX
```

Local mode

Tufts profile

```
#!/bin/bash
```

```
#SBATCH --time=00-48:00:00
```

```
#SBATCH -p batch
```

```
#SBATCH -N 1
```

```
#SBATCH -n 1
```

```
#SBATCH -c 2 ## This is the parent script used for submitting children slurm jobs, 2 cores are enough
```

```
#SBATCH --job-name nf-core
```

```
#SBATCH --output=%x-%J-%u.out
```

```
#SBATCH --error=%x-%J-%u.err
```

```
#SBATCH --mail-type=ALL
```

```
#SBATCH --mail-user=XXX@tufts.edu
```

```
module load nf-core/2.10
```

```
export NXF_SINGULARITY_CACHEDIR=/cluster/tufts/biocontainers/nf-core/singularity-images
```

```
nextflow run /cluster/tufts/biocontainers/nf-core/pipelines/nf-core-rnaseq/3.14.0/3_14_0/ \
```

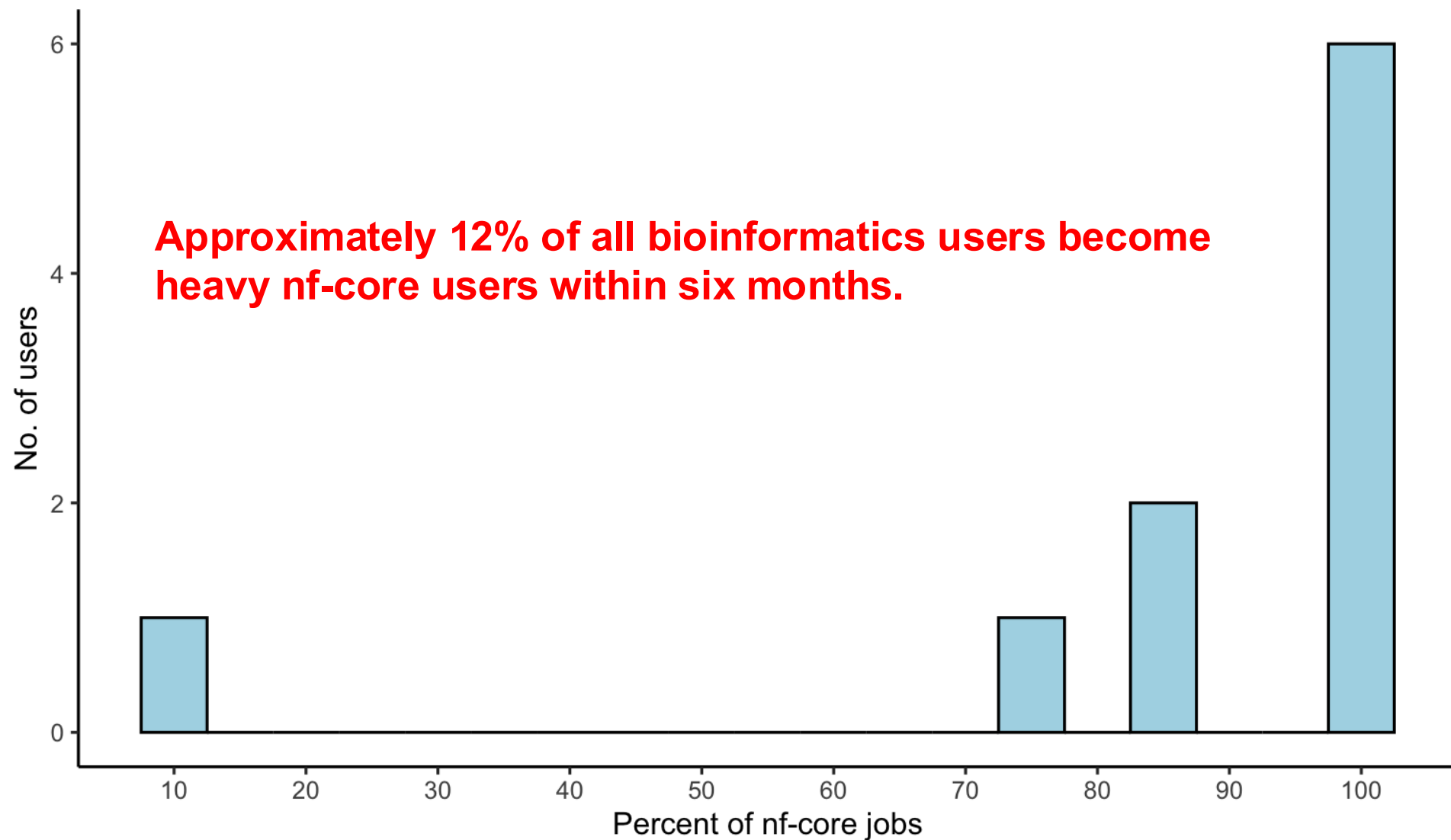
```
    --input samplesheet.csv --outdir output \
```

```
    --fasta ref.fasta --gtf ref.gtf \
```

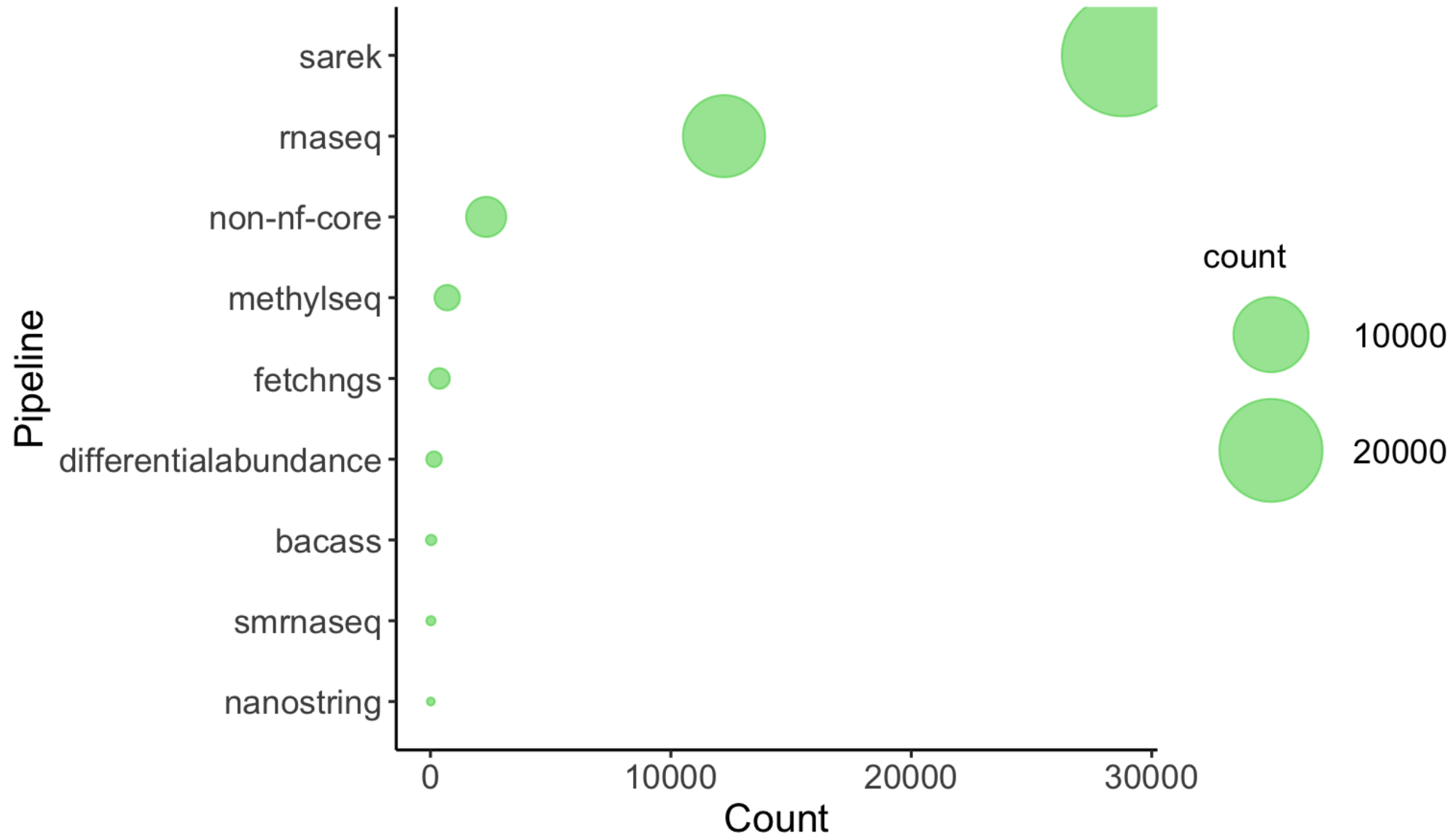
```
    --aligner star_salmon \
```

```
-profile tufts
```

Percent of nf-core jobs among all completed slurm jobs



Usage of nextflow/nf-core pipelines on Tufts HPC



OpenComposer from RIKEN

Open Composer: Web application to generate and submit batch job scripts

Announcements



mnakao Masahiro Nakao

1

Jan 21

Allow me to introduce Open Composer, a web application we are developing. Open Composer operates on Open OnDemand as a Passenger application.

Open Composer is designed to simplify batch job submissions, a core task for HPC cluster users. It offers job management capabilities, including status monitoring and job deletion, promotes parameter reuse, and enables seamless launching of related Open OnDemand applications with a single click.

For more information, please visit:

- GitHub Repository

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Poster

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Top

Application

History

Open OnDemand

FDS

FDS (Fire Dynamics Simulator) is a large-eddy simulation code.

Script Location*

/home/rcs-aot/a03011/

Select Path

Script Name*

Job sh

Job Name

Resource group

small

large

Nodes (1 - 384)*

Procs (1 - 18,432)*

Threads (1 - 48)*

1

1

1

Nodes x 48 => Procs x Threads

Maximum run hours (0 - 72)*

Maximum run minutes (0 - 59)*

1

0

Show advanced option

FDS version*

6.8.0

6.7.9

6.7.8

6.7.7

Input file*

/home/rcs-aot/a03011/

Select Path

Script Contents

```
#!/usr/bin/env bash
#PJM-L "rscgrp:small"
#PJM-L "nodes=1"
#PJM-L "procs=1"
#PJM-L "elapsed=1:0:00"
export OMP_NUM_THREADS=1

./vol0004/apps/oss/spack/share/spack/setup-env.sh
spack load fds@6.8.0

cd /home/rcs-aot/a03011
mpirun fds_mpi_fugaku
```

Submit

Open Composer version: 1.0.0

This application was inspired by discussions on the [Open OnDemand discourse](#). It dynamically generates batch job scripts, similar to BYU Job Script Generator, and streamlines web form creation for various applications. similar to Drona Composer.

Jan 21

1 / 21

Jan 21

2h ago



AlphaFold3 in OpenComposer

Top

Application

History

Open OnDemand

AlphaFold3

This application generates a job script in Slurm.

Script Location*

/cluster/home/yzhang85

Select Path

Script Name*

job.sh

Job Name

Partition

gpu

Number of cores (1 - 128)

1

Memory (up to 800GB)

4

GPU Arch

P100

gpu partition only has p100 and a100

Number of GPUs (1 - 8)

1

Max run time hours (0 - 168)

1

Max run minutes (0 - 59)

0

Software Version

3.0.0

json_path

/cluster/tufts/biocontainers/tests/alphafold3/a100_json/2pv7/2pv7_data.json

Select Path

Path to a single input JSON file.

input_dir

Select Path

Path to the directory containing input JSON files.

output_dir*

/cluster/home/yzhang85/demo/

Select Path

model_dir*

/cluster/tufts/biocontainers/datasets/alphafold3/20241219/

Select Path

Users needs to obtain their own copy of the trained model parameters by completing [this form](#).

max_template_date

2021-09-30

Script Content

```
#!/bin/bash
#SBATCH -p gpu
#SBATCH -n 1
#SBATCH --mem 4G
#SBATCH -t 1:0:00
#SBATCH --gres=gpu:p100:1
#SBATCH --error=%x-%J-%u.err
#SBATCH --output=%x-%J-%u.out

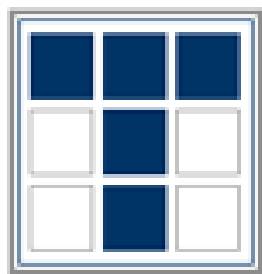
module load alphafold3/3.0.0
DB="/cluster/tufts/biocontainers/datasets/alphafold3/20241219/public_databases"

run_alphafold.py --output_dir="/cluster/home/yzhang85/demo/" \
  --json_path=/cluster/tufts/biocontainers/tests/alphafold3/a100_json/2pv7/2pv7_data.json \
  --db_dir="$DB" \
  --model_dir="/cluster/tufts/biocontainers/datasets/alphafold3/20241219/" \
  --max_template_date="2021-09-30" \
```

Thank you!



McHPCC



TechSquare Inc.
Morgan Ludwig



Tufts TTS Research Technology

Patrick Florance

Jay McGlothlin

Delilah Maloney

Kyle Monahan

Xue (Shirley) Li

...